Ritesh Somashekar

AI & ML Engineer

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PROFESSIONAL SUMMARY

- **AI & ML Engineer** with **5+ years** of experience growing from **data engineering** to **machine learning** and **Generative AI**, now building **RAG systems** and **LLM applications**; published at **ACM WSDM '26** and **IEEE PST 2026**.
- Built healthcare data and ML pipelines at **Carelon Global Solutions**, then led RAG data, retrieval, and validation work at **George Mason University** using **Python, LangChain, LlamaIndex, Llama 3, ChromaDB, PyTorch,** and **AWS**.
- Proven results: **RAG recall@10 raised from 61% to 88%**, a **3.0% LLM hallucination rate**, **99.9% pipeline reliability**, and **60% faster model training**.

TECHNICAL SKILLS

Programming: Python, SQL, R, Java

Generative AI & RAG: LLMs, Retrieval-Augmented Generation (RAG), LangChain, LlamaIndex, Prompt Engineering, Embeddings, Vector Databases (ChromaDB, Pinecone, pgvector, FAISS), GPT-4, Azure OpenAI, Llama 3, Ollama, Agentic AI, LoRA/PEFT

LLM Evaluation: RAGAS, Recall@K, Hallucination Detection, Guardrails, Ablation Studies, A/B Testing

Machine Learning & NLP: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, BERT, XGBoost, LightGBM, SHAP, Anomaly Detection, Time Series Forecasting

MLOps & LLMOps: MLflow, Model Monitoring, Drift Detection, Docker, Kubernetes, FastAPI, CI/CD (Jenkins), Git, DVC

Data Engineering: Apache Airflow, Apache Spark, PySpark, Kafka, Databricks, Snowflake, Delta Lake, Hadoop, SQL Server, MongoDB

Cloud & BI: AWS (S3, Lambda, Glue, SageMaker, SQS), Azure, Tableau, Power BI

EDUCATION

Master's in Data Analytics and Engineering | George Mason University, Fairfax, VA, USA | GPA: 3.93/4.0

Bachelor's in Information Science and Engineering | Visvesvaraya Technological University, Bengaluru, India

PROFESSIONAL EXPERIENCE

George Mason University | ***AI & ML Research Associate*** **Fairfax, VA | August 2024 - Present**

- To scale 9-1-1 call-taker training beyond instructor role-play, co-developed **SHIELD** with **Fairfax County DPSC**, an **LLM-agent** platform that simulates calls and coaches trainees in real time; published at **ACM WSDM '26**.
- To generate safe training calls without commercial APIs, co-developed **MATE-911**, a **RAG** system on **Llama 3.1 8B** that achieved **0.81 protocol F1** and a **3.0% hallucination rate**; published at **IEEE PST 2026**.
- Facing unstructured, abbreviation-heavy call records, led the **RAG knowledge base and data pipeline** (OCR, cleaning, chunking) for **160 gold-standard scenarios**, cutting hallucinations from **13.0% to 3.0%**.
- To ground generation in verified protocols, built a **semantic search** retrieval layer with **Sentence Transformers**, **ChromaDB**, and metadata-filtered **top-K retrieval**, raising the pass rate from **49.7% to 97.6%**.
- To support live training sessions, engineered **prompt and context assembly** for **Llama 3.1 8B (Ollama)**, delivering responses in under **1 second** for text and **1.3 seconds** for voice.
- To keep unsafe content away from trainees, led the **validation layer** with **ontology-based guardrails** and **NLI hallucination checks** against NENA protocols, lowering hallucinations from **9.5% to 3.0%**.
- To speed up literature search, built a **RAG research assistant** with **LlamaIndex, pgvector,** and **FAISS**, raising **recall@10 from 61% to 88%** (RAGAS-evaluated) for **40+ researchers** at **190 ms p95** latency.
- To improve answer accuracy across LLM providers, built **LangChain** RAG pipelines with **FAISS** grounding **GPT-4, Claude,** and **Gemini** in verified sources, raising accuracy by **16%** and cutting hallucinations by **19%**.
- To make model decisions explainable, applied **SHAP** to intent classification and behavioral **anomaly detection** models, tuning **LightGBM** alert thresholds to cut false positives by **25%**.
- To run LLMs on limited hardware, fine-tuned **Llama 3** with **LoRA/PEFT** and **quantization-aware training**, shrinking models by **40%** while staying within **3%** of full-precision accuracy.

- To move prototypes into production, set up **MLflow** tracking, **FastAPI** services, **Docker**, and **CI/CD**, cutting release cycles from **2 weeks to 5 days** with **A/B testing**.
- To study community coping during disasters, applied **graph analytics** and **ensemble models** on the **ORC Hopper HPC cluster**, forecasting trends with an **RMSE of 0.89**.

Carelon Global Solutions (Elevance Health) | ***Machine Learning Engineer*** **Bengaluru, India | May 2022 - August 2023**

- To automate clinical text review, built **BERT/BioBERT** entity-extraction models on unstructured medical notes, reaching **80% accuracy** and cutting processing time by **55%**.
- To speed up member support, built a **Generative AI** assistant with **Azure OpenAI**, **LangChain**, and **Pinecone** RAG for **15K+ daily queries**, cutting resolution time by **38%** and raising ticket deflection by **25%**.
- To improve prediction quality, tuned **Scikit-learn** and **XGBoost** models through feature engineering and hyperparameter search, raising accuracy by **8%** and cutting training time by **15%**.
- To serve models reliably, containerized inference services with **Docker, Kubernetes,** and **Jenkins CI/CD**, improving uptime by **20%** and enabling real-time EHR integration via **REST APIs**.
- To keep production models accurate over time, set up **model monitoring**, **drift detection**, and automated retraining pipelines, sustaining **95%+ accuracy** for **12 months** across **200+ end users**.
- To speed up model training, built ingestion of **400K+ XML/JSON healthcare records** into **Snowflake** and **Databricks**, accelerating training by **60%** and cutting preprocessing time by **70%**.
- To fix inaccurate provider data feeding ML models, automated daily **NPPES** ingestion in **Python**, improving data accuracy by **74%** and reducing processing overhead by **60%**.
- To support ML at scale, designed **20+ ETL pipelines** with **Airflow, PySpark,** and **AWS (S3, Lambda, Glue)**, raising throughput by **70%** and reaching **99.9% reliability** under **HIPAA**.

Carelon Global Solutions (Elevance Health) | ***Data Engineer*** **Bengaluru, India | October 2020 - April 2022**

- To reduce manual claims reconciliation, redesigned **claims-balancing ETL workflows** in **Python** and **SQL**, cutting operational overhead by **60%**.
- To eliminate costly invoice errors, automated **medical invoice processing** pipelines with rule-based classification, reducing manual errors by **70%**.
- To give business teams faster insights, delivered self-service **Tableau** and **Power BI** reporting pipelines, cutting reporting latency by **40%**.
- To handle growing data volumes, migrated **Hive** and **Hadoop** workloads to **Apache Spark** and **Azure Data Services**, improving processing speed and scalability.

PROJECTS

Agentic AI Ticket Automation System | SecureGPT, LLaMA, AWS SQS, JIRA

- To cut security-ticket triage time, built an **agentic AI** system with **SecureGPT** and **LLaMA** on **AWS SQS** that classifies threats and auto-creates **JIRA** tickets, reaching **93% accuracy** and **70%** faster triage.

PUBLICATIONS

- R. S. Nayaka, **R. Somashekar**, K. B. Laskey, L. Wells II, H. Purohit. "SHIELD: A Framework for Online Conversational AI System for Self-Regulated Learning at Emergency Communications Centers." *ACM WSDM '26*. doi:10.1145/3773966.3779411
- R. S. Nayaka, **R. Somashekar**, H. Purohit. "MATE-911: Multimodal AI-powered Training Environment for Next-Generation 911 Dispatcher Preparedness." *IEEE PST 2026*.